

Climate change and its impact on natural resources

Jyoti Kushawaha¹, Sivaiah Borra², Abhishek Kumar Kushawaha³,
Gurudatta Singh⁴ and Pardeep Singh⁵

¹*School of Environmental Sciences, JNU, New Delhi, India,* ²*Indian National Centre for Ocean Information Services (INCOIS), Hyderabad, India,* ³*KIET Group of Institutions, Ghaziabad, India,*

⁴*Institute of Environment and Sustainable Development, Banaras Hindu University, Varanasi, India,*

⁵*Department of Environmental Studies, PGDAV College, University of Delhi, New Delhi, India*

14.1 Introduction

Climate change was the natural phenomenon occurring even before. Still, since a few decades ago, its outbreak becomes severe and shows adverse effects on nature and life. Increasing population and industrialization releases toxic greenhouse gases into the atmosphere and changes hydrological and weather pattern throughout the world. At some places, drought persists for a longer time, and at some places, massive flooding occurs due to precipitation (IPCC, 2007b). The disappearance of glaciers, valleys, and snowcaps affects the downstream habitats adversely. These changes possess severe stress on natural resources like less availability of water resources for agriculture, energy production, domestic uses and ecosystems all around the world. Climate change is one of the biggest threats for the environmental, social, and economic perspective, which is a major challenge for sustainable development, mainly in developing countries. Climate change often results in water scarcity, land degradation, deforestation, and several health issues (Oeurng et al., 2019). The variability of climate change, government policies, and their responses which are linked with socioeconomic development affect the country's sustainable growth (Dhar and Mazumdar, 2009).

Climate change increases global warming and considered as one of the most discussed topics of political controversy worldwide. The world has gone with several debates in the past to confirm whether global warming is real or just a hypothetical thought, but now the effects are quite visible. The significant change in earth's climate brings a massive disruption on every segment of the world's activity as well as a natural phenomenon. Some of the scientific theories suggest adverse repercussion of climate change enhanced by global warming. Some of them are inundation of coastal cities, drying of agricultural fields, melting of glaciers, increasing diseases,

sea-level rise, change of pattern in ocean currents, and other disasters. Some of the crucial indicators of ongoing climate change are:

- temperature,
- the sea-level growth and submergence of the coastal villages and lands,
- the retreat of the major and minor glacial layer,
- extinction of the biodiversity, and
- frequent forest fire, etc.

In the present scenario, the consequences can be seen in the form of heatwaves and associated droughts, extreme and frequent floods, tropical storms, cyclones, sea-level rise, etc. have already affected livelihood across the world.

Global warming increases the average temperature of the earth such as its surface, ocean, and atmosphere. This phenomenon started years ago at a controlled rate. In the past, small climate changes had significant impacts on human activities like agricultural practices, traceroutes for new markets, etc. In the case of spreading human civilization, warming has been proved as a favorable point. After industrial revolutions, the rate of global warming becomes very high due to climate change. According to the World Meteorological Organization (WMO), the past two decades were the warmest years.

The rate of earth's climate change has increased rapidly in an unprecedented manner in recent past years, making the world much warmer. According to the temperature analysis report of WMO, the state of global change in 2019 has ended with the average global temperature of 1.1°C more than the estimated preindustrial average (WMO, 2020). In the previous decade, 2019 was the warmest year without El-Nino effect after 2016, representing as an indicator of the increased global temperature phenomenon. Since 1880, the average global temperature has risen by more than 1 degree. Another alarming point is, the two-third of total warming has occurred after 1975 by the rate of around 0.15°C – 0.20°C per decade. This can be termed as a worrisome situation because the one-degree increase of average temperature has enough heat to warm all oceans, atmosphere, and earth's surface significantly. The average temperature data is based on two factors, that is, how much energy the planet receive through sun and how much energy it radiates back to space. As we all know, that second factor is directly proportional to the availability of greenhouse gases in the atmosphere that is heat-trapping in nature and affects the change in temperature. Greenhouse gases are most important for life on earth because they trap the heat radiated back to the space from earth's surface in the atmosphere and keep the earth warmer and much comfortable for survival. In the last few decades, the chemical composition of greenhouse gases of the atmosphere has been disturbed due to anthropogenic activities. According to the report of WMO (2020), in comparison to the preindustrial era, the level of CO_2 , CH_4 , and N_2O has increased by 147%, 259%, and 123%, respectively.

The global change arises from natural resources extraction and dumping huge tons of waste disposal, which is directly associated with drastic population increase and economic growth. The development race has pumped up the race over industrialization. Large numbers of industries, factories, machines, vehicles etc. are consuming various forms of natural resources and releasing harmful gases like CO₂, CO, N₂O, and CH₄. The gases interfere with the atmosphere's normal chemical composition as well as its average temperature, consequently affecting the planet. The effect of global warming can be observed significantly around the poles. Its impacts not only affect climate and environment but almost every segment of life such as economic, social, and cultural. The rate of greenhouse gases emission is high at all time in today's world. If the rate continues, the scenario will be much more deteriorating and also impossible to hold global warming below 2°C ([World Wildlife Fund, 2016](#)).

Global warming has the potential to cause such drought in various regions that can completely disturb the balance of life. The rising temperature has the effect of drought and habitat fragmentation. Still, both are severe stressors for water resources, especially for groundwater ([Woodward et al., 2010](#)).

14.2 Water resources

One of the most critical resources is water circulating throughout the biosphere via the hydrological cycle that sustains the life functions properly. The hydrological cycle also flows the nutrient cycle that maintains life on earth. The global temperature rise results in the rapid melting of glaciers and icebergs. The higher evaporation rate ultimately results in extra precipitation which affects the groundwater recharge pattern. The change of precipitation, melting of snow and icebergs, warming of permafrost at higher regions are directly linked with heavy runoff and downstream water communities that alter hydrological water system.

The climate change has various type of effects, it can bring drought condition in one geographical area and flood like in another. These conditions ultimately affect primary productivity, adversely affecting the lives of poor people and reducing access to food. Among all the other effects, the rise in temperature will lead to enhanced water demand by humans and animals as severe heatwaves and drought deplete the quality of water and also reduce the availability. The lakes, rivers, and aquifers are facing water-stressed conditions and also getting polluted day by day. More than 50% of the wetlands have been dried up. According to the report of [India Water Aid \(2019\)](#), the population exposed to heatwaves has been increased to around 40 million in five years (2012–2016) in India. The competition due to water will adversely affect the other necessary human activities like energy production, food crop production, industrial production as well as domestic consumption.

The temperature rise will put a significant effect on our daily life through hydrological means. This will negatively affect the sparse population, mainly in countries like India.

India, which has 17% world population and only 4% of total water resource, will face severe water scarcity. Our country has 12 major river systems interwoven to meet the primary demand of water. Around 85% of available water is utilized for agriculture, 8% for domestic use, and 5% for industrial utilization ([Aquastats, 2003](#)). The southeast Asian countries like China, Nepal, Bangladesh, India, and Pakistan use nearly half the total groundwater available ([Rivière and CD, 2015](#)). They face water stress condition and its negative effect on the population as well as the economy of the country ([Lohani and Ait-Kadi, 2013](#)). Several studies are present on the degradation of water resources and provide shocking results in different geographical regions. Mediterranean basins are very vulnerable to global climate change in comparison to Alpine and continental basins. According to the Study of [Jorda-Capdevila et al., 2019](#), the Ebro in Mediterranean basin faces significant decrease of up to 40% in water availability.

The situation is infrequent in comparison to Alpine and continental basins. Around 224 million population of India have less availability to safe drinking water ([India Water Aid, 2019](#)). Freshwaters are present in the form of rivers, aquifers, and streams. India has a network of river system having the input of two primary sources, that is, Himalayan glacial melt and rainfall. Around 80% of the river flow only during the southwest monsoon season, and they are seasonal rivers ([FAO, 2015](#)). After industrialization, the heavily polluted surface water becomes unworthy for drinking purpose; therefore, people switch to groundwater to meet their needs. In India, the agriculture sector covers a significant proportion. Around 3/5 of the population depend on agriculture to sustain their livelihood. Agriculture requires a very high percentage of freshwater for irrigation. Depending on the origin of water, irrigation has been done by canals, tanks, groundwater wells etc. as shown in [Table 14.1](#).

According to the report of [FAO \(2015\)](#), it becomes clear that groundwater consumption for irrigation purposes has increased considerably.

14.2.1 Distribution of water

Distribution of water in nature is extremely irregular. Out of the total water availability, only 3% is available as freshwater that can be utilized by humans whereas remaining 97% are present in the form of marine or ocean that cannot be used because of its saline nature.

Table 14.1: Different types of irrigation and its percentage use in different years.

Irrigation by origin	1993 (%)	2001 (%)
Canal	34	29
Tank	6.5	4
Groundwater	53	63
Undefined sources	6.5	4

Courtesy: [FAO, 2015](#), AQUASTAT Country Profile - India.

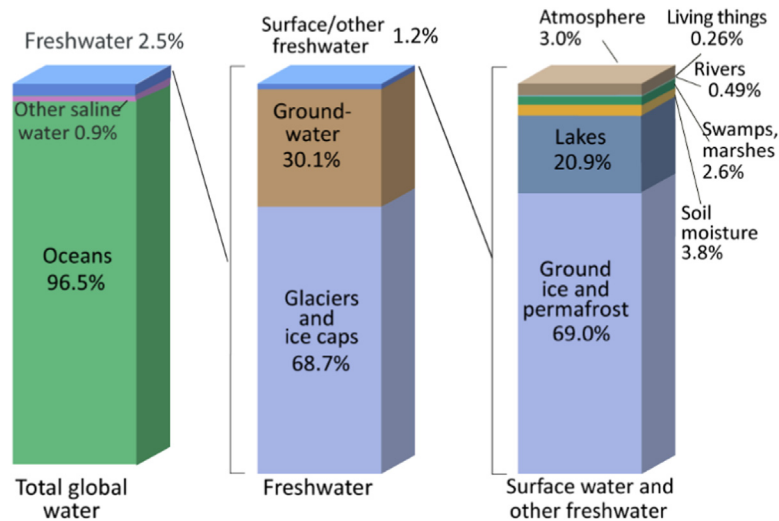


Figure 14.1: Distribution of water on the earth surface.

Source: Adopted from [Gleick, 1993](#), *Water and conflict: fresh water resources and international security. Int. security*, 18(1), pp. 79–112.

The distribution of the freshwater as mentioned above as a whole is 69% in glaciers, 30% in aquifers, and less than 1% in lakes, rivers, and swamps in the form of running water. Only one per cent of the superficial water present on this earth is utilized by humans. At the same time, 99% of that utilizable quantity is present in the underground as groundwater in aquifers (USGS). The water distribution on the earth is presented in [Fig. 14.1](#) and distribution of water resource by volume based on all sources are presented in [Table 14.2](#).

14.2.2 Climatic conditions and water scarcity

According to [IPCC \(2007a\)](#) report, more than half of the world has been facing the adverse impact of climate change over water resources. Significant stress is on water resources in the present changing climatic regime, especially in those countries where more than 60% of the population depends on precipitation for their agriculture. In India, the monsoon pattern varies randomly, and even slight delay in monsoon might cause the food scarcity due to imbalance in hydrological resources ([Mas-Pla and Menció, 2019](#)). Water scarcity has increased due to various contaminations. Drying of rivers and aquifer system has also reduced the dilution effect, enhancing the disease of surface and groundwater resources. Besides, different geological, geographical, and economic regions have produced and added their own respective contaminant in the water system. Some of the examples are nitrate contamination in the highly fertilized agriculture area, the release of arsenic in reducing condition, saline water intrusion in the coastal regions due to overexploitation of groundwater, fluoride contamination in fluoride bearing geological regions, etc.

Table 14.2: Distribution of water resource by volume on the basis of all sources.

Source of water	Volume of water in km ³ (cubic miles)	Percentage of water	Percentage of total freshwater
Oceans	1,338,000,000 (321,000,000)	96.5	—
Pacific Ocean	669,880,000 (160,710,000)	48.3	—
Atlantic Ocean	310,410,900 (74,471,500)	22.4	—
Indian Ocean	264,000,000 (63,000,000)	19.0	—
Southern Ocean	71,800,000 (17,200,000)	5.18	—
Arctic Ocean	18,750,000 (4,500,000)	1.35	—
Ice and snow	24,364,000 (5,845,000)	69.6	1.76
Glacier	27064000 (5,773,000)	68.7	1.74
Antarctic ice sheet	2340000 (5,600,000)	6.68	0.17
Arctic islands	83,500 (20,000)	0.24	0.006
Mountain ranges	40,600 (9700)	0.12	0.003
Ground ice and permafrost	300,000 (72,000)	0.86	0.022
Fresh ground water	10,530,000 (2,530,000)	30.1	0.76
Soil moisture	16,500 (4000)	0.047	0.0012
River	2120 (510)	0.0061	0.00015
Swamps	11,470 (2750)	0.033	0.00083
Biological water	1120 (270)	0.0032	0.000081
Atmosphere	12,900 (3100)	0.037	0.00093
Other fresh water lake	15,200 (3600)	0.043	0.0011
Fresh water lake	91,000 (22,000)	0.26	0.0066

14.3 Ocean resources and climate change effect

More than 70 out of 100 part of the earth's surface is shielded by ocean. From reducing the extremes of climate to producing the oxygen we breathe, from the food we eat to the excess carbon dioxide we produce, several human communities have been served by these oceans. The enhanced greenhouse gas emission affects coastal and marine communities by altering sea temperatures and melting ice, changing ocean currents, weather patterns and sea levels.

Extreme weather events surely get affected by ocean warming. For example, as oceanic hurricanes are developed, the sea temperature is increased. Since heat is an energy, it can enhance the catastrophic and destructive power (Webster et al., 2005; Hoyos et al., 2006). It is still not clear whether the frequency of cyclone occurrence has been affected by climate change or not, however, their magnitude is likely to intensify further (IPCC, 2007b). The rising intensity of cyclones threatens not only people but also marine and coastal ecosystems.

According to reports, the upcoming centuries will witness the sea-level rise (SLR), although the magnitude of the increase remains indeterminate and uncertain (IPCC, 2014, Kopp et al., 2014; Horton et al., 2014). The global average rate of SLR reported 1.8 mm/year in the interval of 1961 to 2003. In contrast, the pace was faster from 1993 to 2003, about

3.1 mm/year. The sea-level rise adversely affects salt marshes, mangroves, coastal aquifers and the coastal community (IPCC, 2007b). The Fifth Assessment Report on Climate Change (AR5) of the Intergovernmental Panel on Climate Change (IPCC) predicted a “chance (> 66% probability)” in the world-average SLR of 52–98 cm by 2100 as compared to 1986–2005 [Representative Concentration Path (RCP) 8.5 Scenario (Riahi et al., 2011)]. It indicated a world-average SLR of 1 to >3 m by 2300 with Medium Confidence (IPCC, 2014). Uncertainties in ocean currents are triggered by climate change which can lead to significant changes in local climate, climate patterns and future human migration. According to the report of Rahmstorf (2006), the thermohaline circulation downfall has been broadly discussed as one of the “low probability-high impact” risks concomitant with global warming.

In respect to forthcoming warming of the climate and surface-ocean waters (IPCC, 2014), it is equitable to presume that the climate system will create more recurrent and extreme tropical cyclones in various dimensions and trajectories (Villarini and Vecchi, 2013; Lin et al., 2016). The magnitude of tropical cyclone features remains uncertain (Lin, et al., 2012, Horton et al., 2015). The increase from the US \$6 billion in 2005 to \$1 trillion by 2050 has been estimated as the average annual cost of flood damage in coastal cities, at current levels of protection (Hallegatte et al., 2013). Nearby glaciers and sea icebergs also shakes the climate. Most of the incoming sunlight energy is reflected away from the ice surface (Curry et al., 1995). Warming of Arctic shrinks the sea ice cover and the dimmer open water substitutes the sea ice. Thus, generating a response of the heating and ice melting (Curry et al., 1995; Anisimov et al., 2001).

On the other hand, the three significant threats to the health of coral reefs have been affected by climate change. This includes the intake of atmospheric carbon dioxide by sea, increased storms and warming waters. Marine intake of CO₂ leads to more acidic water that negatively interferes with corals in maintaining and forming their shells. Damaging currents of winds can destroy corals. Warming waters initiate bleaching and many diseases in corals (Hughes et al., 2003). In 2016, the Great Barrier Reef experienced very high marine heatwave. The study narrows the gap between theory and practice. It examines the risks of ecosystem breakdown to investigate future-warming events affecting coral reef communities. They identify the main driver, define various stages and also set quantitative fall limits (Hughes et al., 2018).

Marine ecosystem and its species become vulnerable due to ocean acidification. It diminishes the capability of maritime organisms like plankton, corals, etc., in building their shells and skeletal morphology. It enhanced prevailing physiological stresses and hampered the physical and biological functions respiration and reproduction and affects the growth and survival rates of some species in their early life stages (IUCN, 2017a, b). Impact of climate change on ocean resource has been shown in Fig. 14.2.

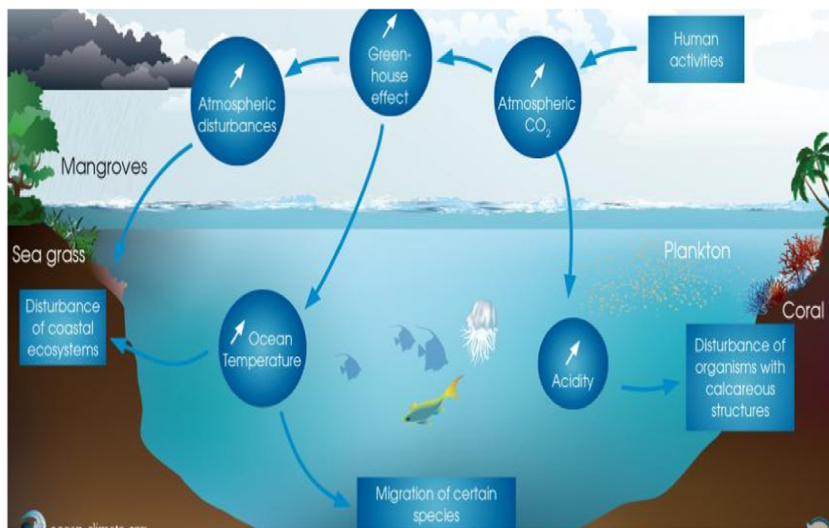


Figure 14.2: Impact of climate change on ocean resource.

Source: Adopted from *IUCN, 2017a, Deforestation and forest degradation*; *2017b, The ocean and climate change*.

14.4 Forest resources and vegetation

Nature provides so many essential resources for human life. Still, the slight change in temperature can affect these resources and ultimately the human life. Another most valuable resource that can be significantly affected by the rise in temperature is the forest resources. The consequences of the climate change has a wide impact on vegetation patterns, forests, and species. Increasing CO₂ boosts forest productivity though it also enhances the drought linked with forest fire, risk for the invasive species. Moreover, the increased temperature disturbs the distribution and composition of tree species and the species variation proceeds northwards. Practically, this can be a transition phase from maple and birch as the preferred tree species to oak and hickory by 2100. Few species like red pine may entirely retreat from the Great Lake region by the end of the century. The climate change impact on forest resource and vegetation are presented in Fig. 14.3.

Most of the people consider the forest as an area with a dense population of trees, but they are more than the trees. Forest is one of the most critical natural resources which is necessary to sustain life on Earth. World's 80% population of the terrestrial biodiversity resides in the forest. It plays a significant role in the water cycle, habitat protection, soil conservation, carbon sequestration etc. and the relation between forest and climate change is robust and directly connected. The global forests absorb 2.4 billion CO₂ per year, out of which 1/3rd from fossil fuel annually (IUCN, 2017a, b). Forests play the two-way role in the carbon cycle as they add CO₂ gas, that is, one of the most critical greenhouse gas when it is overused.

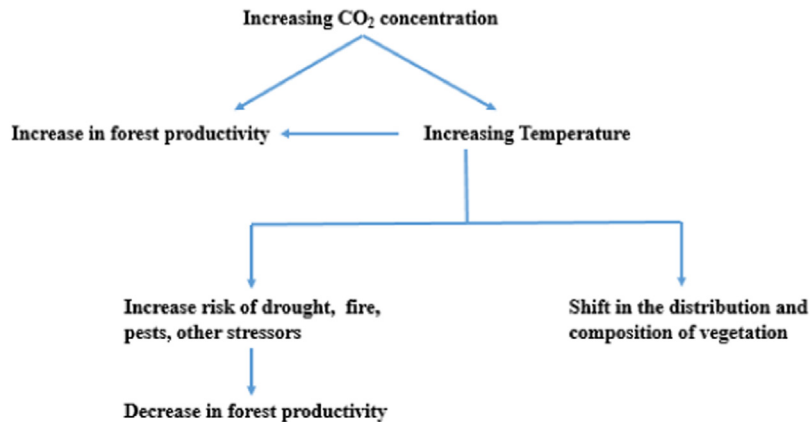


Figure 14.3: Effect of climate change on forest resource and vegetation.

In contrast, they can absorb the CO₂ gas emission by anthropogenic activities or natural resources into their biomass, soil, and products by the process of carbon sequestration. It is a method where trees and forests remove atmospheric CO₂ by converting it into carbon during photosynthesis which is stored in the form of wood and vegetation. Thus acting like “carbon sinks.” Apart from this, it also plays a crucial role in generating adaptability to climate change and providing safety during climatic shocks. These resources are subjected to indiscriminate exploitation and destruction because of anthropogenic activities and climate change. Degradation and forest decline have increased the CO₂ concentration into the atmosphere. Around 10.5 million acres of the tropical forests and peat swamps have been destroyed since the last 25 Years in central Sumatra’s Riau Province (World Wildlife Fund, 2008).

For developing countries like India, it is a very complicated step to achieve the goals of sustainable development, where more than 60% of the population depends on rain-fed agriculture. This agriculture again lacks the appropriate irrigation facilities and forests for their food and services. (Kumar and Kushwaha, 2020) (check this sentence). A study conducted in Uttarakhand, based on the analytical hierarchy process shows that approximately 61% of the forests area is under very high vulnerable category whereas 36% under top vulnerable category (Pokhriyal et al., 2020). The nontimber forest products and its quantity have been declined including fisheries also which is significantly affected by climate change in Amarkantak, Madhya Pradesh. The useful medicinal plants are not available in the present because of forest degradation and forest fire (Ahirvar et al., 2020). The climate change hampers the functionality and structure of the ecosystem and affects its health creating severe consequences.

In the recent era, deforestation has become a common practice to extract various resources. The most common and significant reason for deforestation is converting forest land into agriculture fields and settlement; it happens for years as a part of the economic

development process with an increase in population. Deforestation and forest degradation affects the 1.6 billion people whose livelihoods depend on forests and around a billion of them are the world's most impoverished population (IUCN, 2017a, b). Forest fire is one of the worst situations and also the result of global warming.

In many countries, the lack of an active forest management program and its implementation has been found, creating an imbalance between ecological and economic development processes that ultimately affect the greenhouse gas composition in the environment. As a result, the increase in temperature takes place. Effect of climate change on forest resource and vegetation are presented in Fig. 14.3.

14.5 Fish and wildlife

As habitat zone the respective temperature biodiversity have been shifting northwards and aquatic organism and wildlife as well. The Great Lake becomes an obstacle to migrate of many species and create a less adapting environment to the animals. Wildlife relying on wetlands are at most significant risk, as enhanced evaporation rate reduces total wetland coverage and stress the species residing there. The fish population of the Great Lake region might become less in numbers. The warmer temperature of water leads to the declined coldwater fish population, and they become plentiful. Effect of temperature on fish and wildlife are discussed in Fig. 14.4.

14.5.1 Impact of climate change on freshwater

Climate change effect on freshwater resources varies greatly depending on time and location. The significant effects on freshwater vary from timing, quantity, and types of precipitation. Increase in winter precipitation and decrease in summer precipitation has been

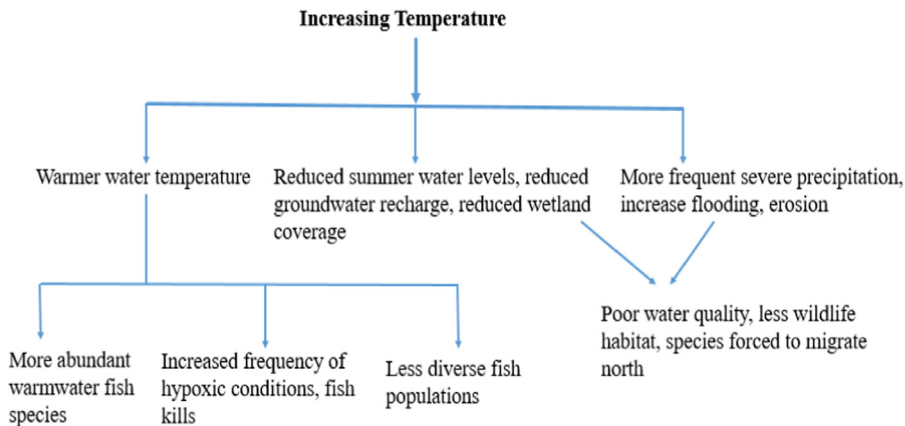


Figure 14.4: Effect of increasing temperature on fish and wildlife.

witnessed in the Pacific Northwest whereas Southwest experiences less precipitation whole year.

According to the report of [Doell et al. \(2015\)](#), the risk of flood and drought might be increased in many areas. Other drivers of climate like increased temperature also has adverse effects on freshwater resources. There are following key impacts are:

1. **Changes in precipitation pattern:** Variation in the timing, quantity, and types of precipitation affects all the form of freshwater including its quantity and quality.
2. **Temperature rise:** Rising temperature accelerates the rate of melting ice, snowpack, and permafrost that contribute to the short term increase in river flow linked to flooding and followed by narrowing of rivers. Elevated temperature cause substantial water loss through evapotranspiration and lowered the oxygen level in water consequently deteriorate the water quality hence the health of the aquatic ecosystem ([Doell et al., 2015](#)).
3. **Extreme weather events and floods:** The intensity and number of heavy precipitation event increase ultimately affecting hydrological condition (e.g., increased flooding, erosion, and runoff) and water quality.
4. **Sea-level rise:** Sea-level rise contributes to coastal flooding as well as salt-water ingression into coastal aquifers and surface waters.
5. **Water scarcity and drought:** Increased temperature and lesser rainfall contribute to water inadequacy and famine in some regions, particularly in the arid areas like Southwest ([U.S. Global Change Research Program, 2014](#)).
6. **Changes in groundwater supply:** Previously discussed impacts on precipitation and hydrology affect the recharge rate, some areas might experience hasty recharge, but water-stressed areas experience a slower rate of aquifer recharge. Availability of groundwater is get affected by an increase in withdrawal of water in these areas ([Meixner et al., 2016](#)).

In the past few decades, freshwater resources unveiled a direct response to temperature variation (temperature increases), atmospheric humidity and precipitation ([Hader et al., 2007](#)). Basically, concerning physiology and phenology of freshwater organism and their habitats. These responses become noticeable with increasing global surface temperature ([Schippers et al., 2004](#)). Impact of climate change is presented in [Fig. 14.5](#).

14.6 Conclusion

To effectively discuss climate change, there is an exigent need to condense increasing greenhouse gas emission and pollution and also to get prepared for the extreme consequences of global warming. Advance and effective policies should be implemented with the collaboration of businesses to reduce carbon emission. Enhanced emission of

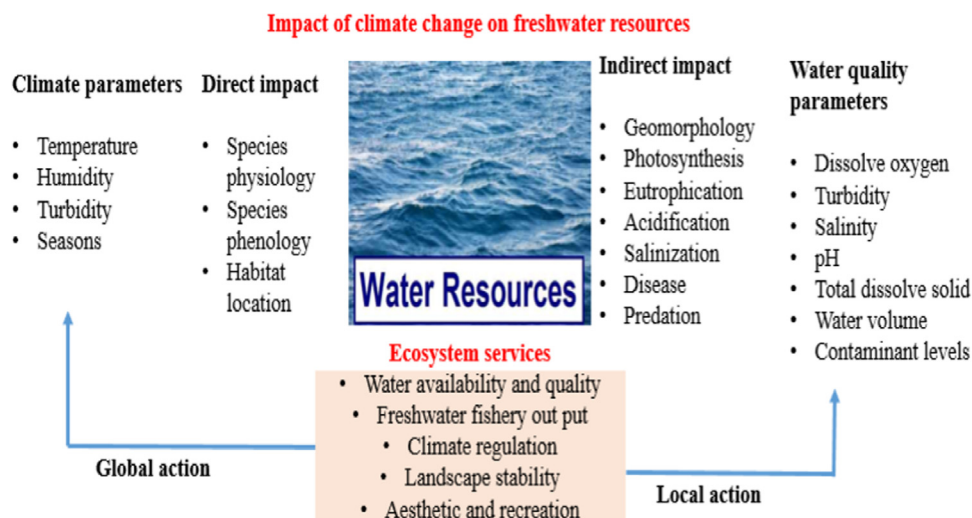


Figure 14.5: Overview of ecosystem services and direct and indirect effect of climate change on freshwater resources.

Source: Modified from Liu et al., 2015, Towards threshold-based management of freshwater ecosystems in the context of climate change. *Ecol. Model.*, 318, pp. 265–274.

natural and anthropogenic greenhouse gases, disrupting natural scenarios require more advance and clean policies to mitigate the impact of climate change on oceans like ocean warming, sea-level rise, uncertainties in ocean currents, global circulation system, coastal flooding, increase in frequency and intensity of tropical cyclones and associated surges, sea ice, and ocean acidification. It is very much necessary to create awareness among people on climate change and its adverse impacts on natural resources and living being.

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